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BSIT 2-1
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DATE

midterm Activity #3 - Basic ATM System

```
package Java Unoj
public class ATMTransactionCounterAbenoja {
    public static void main (String[] args) {

    double balance = 5000.0;
    int count = 0;
    boolean active = true;

        if ( System. console () == null) {
            System.out. println ("Console not available. Run in
                                Terminal.");
            return;
        }

    while (active) {
        System.out. printf("\n ===== ATM Menu ===== ");
        System. out. println( "1. Check Balance\n 2. Deposit \n 3.
            withdraw\n 4. Exit ");
        System. out. print ("Select: ");

        String input = System. console(). readline();

    switch (input) {
        case "1":
            count++;
            System.out. printf("Transaction " + count + ": ");
            System. out. println ("Choice: 1 ");
            System. out. println (" Output: Balance = " + (int)balance);
            break;
```

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```
case "2":
    count ++;
    System.out.println("Transaction" + count + ":");
    System.out.println("Choice: 2");
    System.out.println("Deposit: ");
    double dep = Double.parseDouble(System.console().readLine());
    balance += dep;
    System.out.println("Output: New Balance = " + (int)balance);
    break;

case "3":
    count ++;
    System.out.println("Transaction " + count + ": ");
    System.out.println("Choice: 3");
    System.out.println("Withdraw: ");
    double draw = Double.parseDouble(System.console().
        readline());
    if(withdraw > 2000){
        System.out.println("Output: Exceeds maximum
            withdrawal");
    } else if (withdraw > balance){
        System.out.println("Output: Amount exceeds
            balance");
    } else {
        balance -= withdraw;
        System.out.println("Output: New Balance = " +
            (int) balance);
    }
    break;
```

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```
case "4":
        System.out.println("\n Transaction Count: " + count);
        active = false;
        break;
    default:
        System.out.println("Invalid option:");
    }
  }
}
```

Example outputs:

Transaction 1:

Choice: 1.

Output: Balance = 8000

Transaction 2:

Choice: 2

Deposit: 1500

Output: New Balance = 9500

Transaction 3:

Choice: 3

Withdraw: 1500

Output: Amount exceeds balance

Transaction count: 3

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Explanations:

while loop

- The while loop ensures that the ATM doesn't just run once and then close. When a boolean condition, like active == true, is evaluated, the code block inside is repeated as long as the condition is met. The condition is adjusted to false when the user chooses the exit option.

switch - case

The switch statement "jumps" straight to the correct scenario, such as deposits or withdrawal, after evaluating the number choice. Compared to writing numerous nested-if statements for each option, this approach is much more efficient. In order to prevent the program from acting up, it has a default option for handling invalid inputs.

if-else

Before determining if the user has enough balance to meet the request, it first determines whether the requested amount exceeds 2,000 transaction limit. The "if" or "else if" block cause the user to recieve an error message. The "else" block updates the account balance and performs the subtraction only when all conditions are met.

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THALIA VALIANT